

Understanding Grammar Representation in the Human Brain via Information-Restricted LLMs

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1 Introduction

1.1 Importance of verbs and nouns in language

Ever since the start of elementary school, identifying and distinguishing Grammatical Class (GC) – especially nouns and verbs – plays a fundamental role in the understanding of grammar. Researchers in the field of linguistic neuroscience have long examined whether this distinction was represented in the brain. However, this has proven to be particularly challenging, with many studies showing contradicting results.

1.2 Neuropsychological studies

Early neuropsychological studies conducted experiments with aphasic patients who had strong impairments in the retrieval and production of nouns and verbs respectively. A. R. Damasio and Tranel (1993), for example, observed impairments in noun retrieval for two patients with damage to the *left anterior and middle temporal lobe*, while the *left premotor cortex* was named as a possible cause of the third patient’s impairment in the retrieval of verbs. This observation gave rise to the *fronto-temporal* dichotomy, suggesting that nouns might be processed in temporal areas while verbs are processed in frontal areas.

However, subsequent studies have found contradicting results, with some patients showing frontal damage without verb deficits and temporal damage without noun deficits (H. Damasio et al. 2001; De Renzi and Pellegrino 1995). Additionally, some of the verb deficits could be explained by the difference in imaginability of the words (Crepaldi et al. 2013).

1.3 Psycholinguistic Models

- **Lexicalist:** GC information is automatically retrieved upon retrieving the word.
- **Combinatorial:** GC is viewed as *procedural knowledge*. Nouns/Verb category is treated as computational objects and only relevant for integration tasks.

- **Emergentist:** GC information is a property emerging from semantic constraints like nouns $\rightarrow$ objects and verbs $\rightarrow$ actions.

Another hypothesis for the apparent distinction is that the brain does not process GC *per se* but rather the meaning that is usually associated with the word (Vigliocco et al. 2011). Particularly, nouns are usually associated with **objects** while verbs tend to be associated with **actions**. This hypothesis also falls into the range of the **emergentist** views of psycholinguistics, wherein GC distinction is viewed as an emergent property driven by semantic constraints.

2 Materials and Methods

2.1 Stimuli

The speech stimuli consist of 11 short stories taken from **The Moth Radio Hour**, a storytelling program where true autobiographical stories are told in front of a live audience. These stimuli were used in previous papers (Huth et al., 2016), including the one where the response data was gathered (Deniz et al., 2019). These stories provide a naturalistic stimulus.

Out of these 11 stories, 10 were used as training data while one 10-minute story was used as validation data. The latter was played two times during the experiment (in two separate scans) to which the responses were finally averaged across the two runs.

These stories were then transcribed and stored as Praat’s TextGrid objects, a text-based file format containing the transcription of the words spoken in the stories and the times when they were spoken. Additionally, the text grids also contained phoneme information which was not used in the project.

2.2 fMRI Responses

Each spoken (and written) story was presented during a separate fMRI scan. The length of each scan was the same as the story. The sampling rate was 2 seconds, so $1 \text{ TR} = 2 \text{ s}$. Each scan included 10 seconds (5 TR) of silence both before and after the story. These data were collected during 23-hour scan sessions that were performed on different days. MRI data were collected on a 3T Siemens TIM Trio scanner at the UC Berkeley Brain Imaging Center using a 32-channel Siemens volume coil.

Each functional run was motion-corrected using the FMRIB Linear Image Registration Tool (FLIRT) from FSL 5.0 (Jenkinson and Smith, 2001; Jenkinson et al., 2002). All volumes in the run were then averaged over time to obtain a high-quality template volume. FLIRT was also used to automatically align the template volume for each run with the overall template, which was chosen to be the temporal average of the first functional run for each participant.

The responses were provided as HDF files. For each subject and each modality, there were two files: one for the training stories and one for the validation story, for a total of 36 files. Each training file contains a dictionary with the stories as keys and the BOLD responses as values. Each validation file contains a

dictionary with *story_11* as the key and the responses from both repetitions as values.

The responses are represented as a 2D numpy array with shape $(N_{TR} \times N_{voxels})$. The amount of TRs depends on the story, while the amount of voxels is unique to the subject. The response data also included the 10 seconds of silence before and after the story was played.

2.3 Voxelwise Encoding Model Framework

Voxelwise encoding models (VEM) aim to predict the BOLD response for each individual voxel v from the stimulus S . This is usually done by first extracting one or more sets of features (called feature spaces) from the stimulus, then modeling the brain response as a linear combination of those feature spaces.

This corresponds to approximating the function $f(S) = g(S)\beta$ where g transforms the N -length stimulus into an $N \times P$ feature space with β as the $P \times V$ weights, V being the number of voxels. This can be accomplished by fitting the feature spaces $g(S_{\text{train}})$ to the responses R_{train} of a training dataset, yielding a model $\hat{f}$. The hemodynamic delay is accounted for by including delays of 0, 4, 8, and 16 seconds for each feature, resulting in a finite impulse response (FIR) model.

Finally, the model accuracy is calculated as the per-voxel Pearson’s correlation r_v between the model predictions on a validation dataset $\hat{R}_{\text{test}} = \hat{f}(S)$ and the actual responses R_{test} .

The value of r_v can then be interpreted as the extent to which voxel v ’s response to the stimulus can be explained by the given feature space. Projecting r_v onto the cortical surface then yields a sensitivity map to the specific feature space. This shifts the focus from experimental design to the choice of features, allowing researchers to test different hypotheses utilizing the same dataset by varying the feature space.

Banded Ridge Regression

To fit the multiple feature spaces to the BOLD responses I used Banded Ridge Regression (BRR), which is based on Ridge Regression, a linear regression method that augments the least-squares objective with an ℓ_2 penalty to reduce overfitting to the training data. This corresponds to finding the weight vector

$$\hat{\beta} = \arg \min_{\beta} \|X\beta - y\|_2^2 + \alpha \|\beta\|_2^2$$

with $\alpha > 0$ being the regularization parameter, which is typically selected through grid search or random search with cross-validation.

BRR extends Ridge Regression by assigning a separate regularization parameter α_i to each feature space X_i nunez2019.

$$\hat{\beta} = \arg \min_{\beta} \left\| \sum_{i=1}^m X_i \beta_i - y \right\|_2^2 + \sum_{i=1}^m \alpha_i \|\beta_i\|_2^2$$

This is important since the feature spaces can vary greatly in size and signal-to-noise ratio and thus need different levels of regularization.

2.4 Features

To reproduce the original study, the 985-dimensional Eng1000 features were provided.

To account for response variance caused by various modality-dependent nuisances, several additional features were provided.

Phonemes

A list of phoneme-time pairs was constructed for each of the 39 English phonemes. This list was then down-sampled to the fMRI acquisition rate. The result is a $(39 \times N_{TR})$ array. These features were used when fitting the model to the listening data.

Letters

Similar to the phonemes, a 26-parameter model was created to represent the frequency of each of the 26 letters in the English alphabet that appeared in the written stories. This was also down-sampled to the sampling rate, resulting in a $(26 \times N_{TR})$ array.

Additional Features

To account for the highly variable speech rate both within and across stories, single-feature models were constructed. These count the number of words, number of phonemes, number of letters, and number of story speaker’s pauses that occurred during the acquisition of each fMRI volume (2.0045 s). To account for the variable word lengths during the visual presentation, a single-feature word length variation model was constructed by taking the variance of word lengths that occurred during the acquisition of each fMRI volume.

2.4.1 Contextual Embeddings

The largest issue with the lexical *English1000* embeddings is the lack of contextual information in the feature space. Put differently, the same word will be mapped to the same feature vector regardless of the context it appears in. For example, the word *fine* has a very different meaning in the sentences “I am fine” and “I paid a fine”.

To account for this, I made use of *contextual embeddings*, which, as the name suggests, aim to include contextual information in the word embeddings. Such embeddings were introduced by the Long Short-Term Memory (LSTM) network ELMo (Peters et al. 2018) and refined by the transformer model BERT (Devlin et al. 2019).

Contextual embeddings play a fundamental role in all modern Language Models (LM), and for this thesis I decided to use GPT-2’s embeddings, as it is commonly used in related literature (Schrumpf et al. 2021; Tuckute et al. 2024).

Given an input sequence comprised of the i -th word of a story w_i and its k -length context $C_i = (w_s, \dots, w_{i-1})$ with $s = \max(0, i - k)$, I can generate the contextual embeddings through the following steps.

Tokenization

Before a sentence or word can be embedded into vectors, it needs to be broken up into a set of token IDs. This turns the input sequence into a sequence of tokens $(t_1, \dots, t_L) \in \mathbb{N}^L$.

Initial Embeddings

Upon receiving the token sequence, GPT-2 first converts the tokens into embedding vectors via a *lookup table* with a fixed embedding size $d_{\text{model}} = 768$, then adds positional encodings to generate the initial embeddings. These initial embeddings are non-contextual.

Hidden Layers

The initial embeddings then go through a sequence of layers where they are first weighted using the attention mechanism (Vaswani et al. 2023), then passed through a feed-forward network to enrich them with contextual information from surrounding tokens. This yields a new contextualized embedding of the same dimensionality d_{model} .

Previous studies have found that the middle layers, e.g., Layer 8 for GPT-2 small, are best at predicting brain activity in VEMs (Schrumpf et al. 2021; Jain and Huth 2018), and Layer 8 is therefore used throughout this work.

Averaging

The hidden state $H_8 \in \mathbb{R}^{L \times 768}$ contains the contextualized embeddings for every token in the input sequence. These tokens are then averaged to return w ’s contextualized embedding e_w :

$$e_w = \frac{1}{L} \sum_{i=1}^L h_8^{(i)}$$

Downsampling

After repeating this process for each word in each story, I have an embedding matrix $E_j \in \mathbb{R}^{N_j \times 768}$ with $N_j = |S_j|$ being the number of words in the j -th story. To fit the features to the fMRI data, the contextual embeddings are downsampled onto a feature space $F_j \in \mathbb{R}^{K_j \times 768}$ via a Lanczos filter, with K_j being the number of sample points (TRs) of the j -th story.

The baseline contextualized embeddings are generated using a context length of 20 words for each word in each story.

2.5 Context Masking

The masking of Parts of Speech (POS) was performed automatically using Python’s `spaCy` library. This library first splits an input sequence into a series of word-like tokens based on language-specific rules. For example, *don’t* is split into two tokens *do* and *n’t*. These tokens are then assigned a POS tag that takes their context into consideration.

Given a set of POS tags P , I can generate a boolean mask for each story:

$$M_j = (f_{\text{mask}}(t_{j,1}), \dots, f_{\text{mask}}(t_{j,n})) \quad f_{\text{mask}}(t) = \begin{cases} 0 & \text{POS}(t) \in P \\ 1 & \text{POS}(t) \notin P \end{cases}$$

Since proper nouns and auxiliary verbs perform similar functions to nouns and verbs respectively, they are included in the noun and verb POS categories accordingly.

Attention Mask

To apply the POS masks during the generation of contextual embeddings for a word w , the mask values for w and its context C are first mapped to the GPT-2 tokens, yielding M'_w . They are then used as *attention masks*, indicating which tokens the model should attend to and which should not. The resulting contextual embedding is calculated by averaging only the unmasked hidden states:

$$\bar{e}_w = \frac{1}{L'} \sum_{i=1}^L m_w'^{(i)} h_8^{(i)} \quad L' = \sum_{i=1}^L m_w'^{(i)}$$

Rather than removing masked tokens from the input sequence — which would alter positional encodings and shift the attention context of remaining tokens — attention masking preserves the original sequence structure while selectively excluding the target POS category from the model’s attention computation.

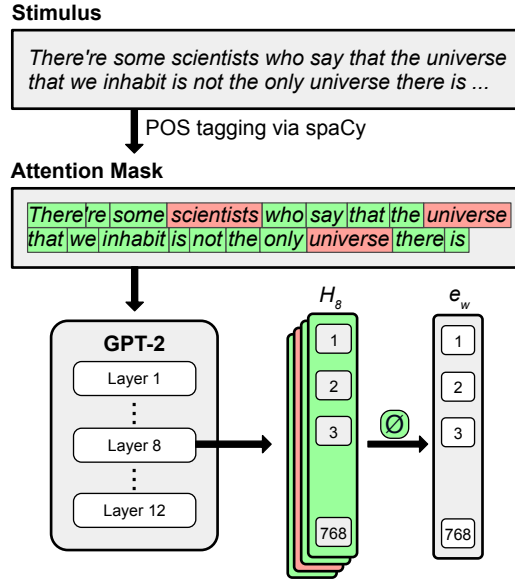


Figure 1. Contextual embedding pipeline with POS attention masking. A stimulus sentence is passed through a noun filter, which uses spaCy POS tags to generate an attention mask (green: unmasked; red: masked). The masked input is fed into GPT-2, whose hidden state at Layer 8, $H_8 \in \mathbb{R}^{L \times 768}$, is then averaged over unmasked tokens to produce a single contextualized embedding $e_w \in \mathbb{R}^{768}$ for the target word.

3 Results

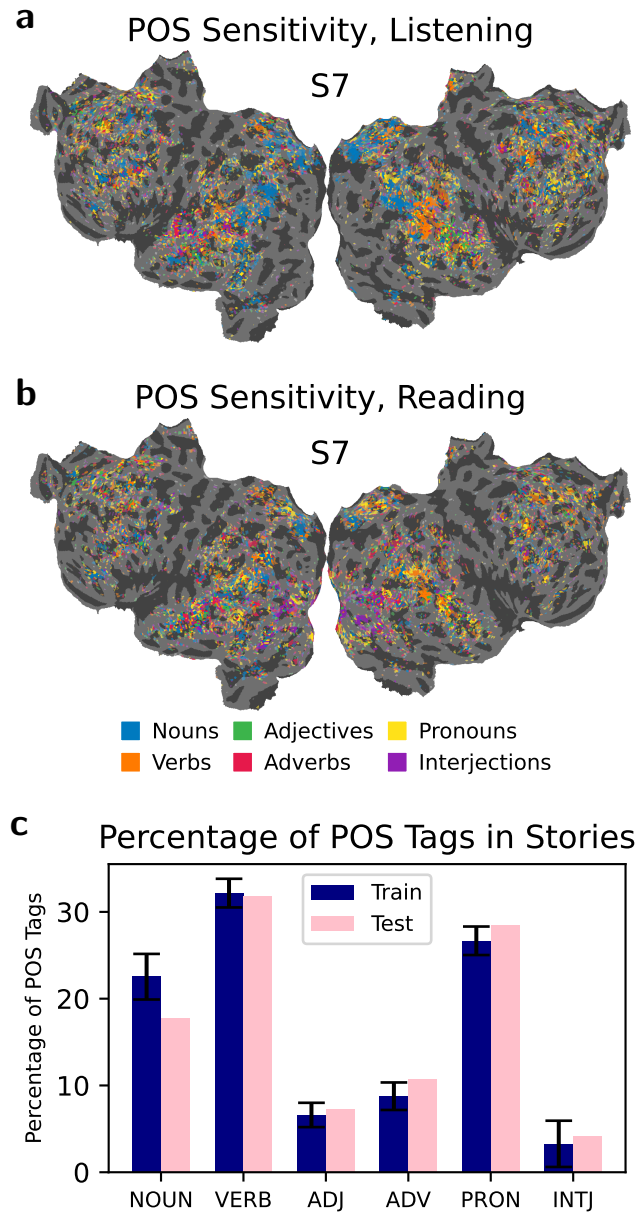


Figure 2. Part of Speech sensitivity across the brain using all POS: nouns, verbs, adjectives, adverbs, pronouns and interjections. **a** Voxelwise model r values in the listening modality using POS-masked contextual embeddings were subtracted from the r values of the baseline contextual model to get the drop in accuracy. The drop in accuracy for each POS was then ranked for each voxel individually and the greatest drop was plotted on the flatmap in the respective color **b** Voxels are colored using the same procedure as in **a** but for the reading modality. **c** Distribution of POS tags in the training stories and the validation story. The percentage of each POS tag was calculated by counting the number of words with that tag and dividing it by the total number of words in the story. The average percentage across the 10 training stories is plotted on the left, while the percentage for the validation story is plotted on the right.

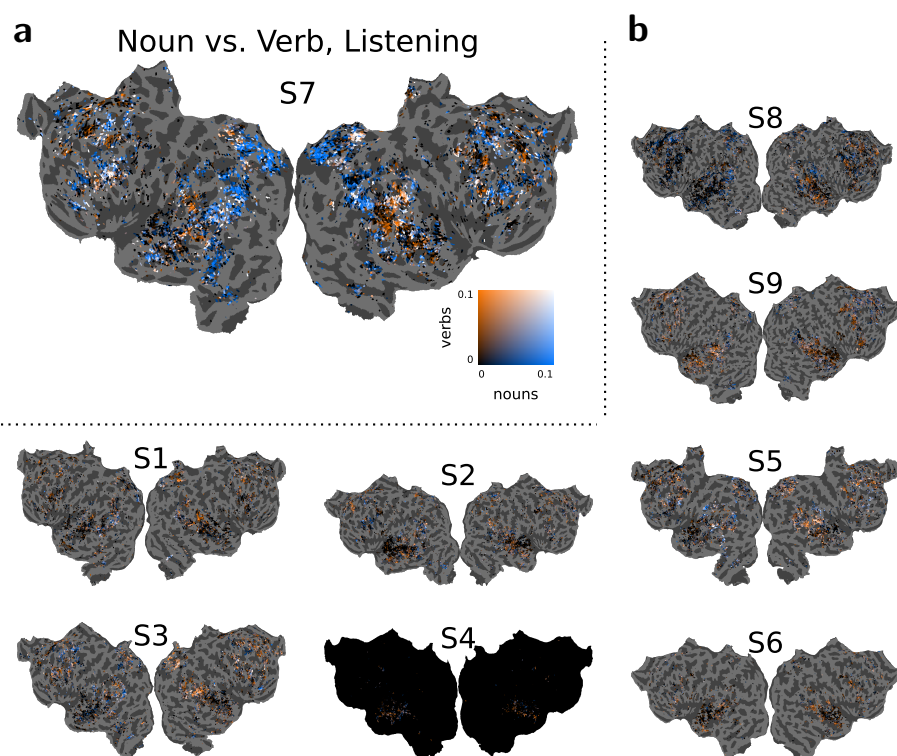
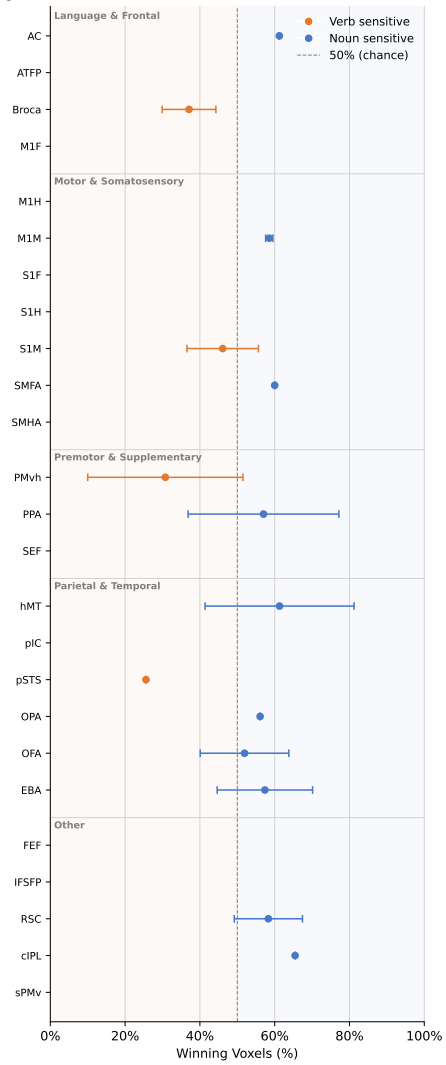
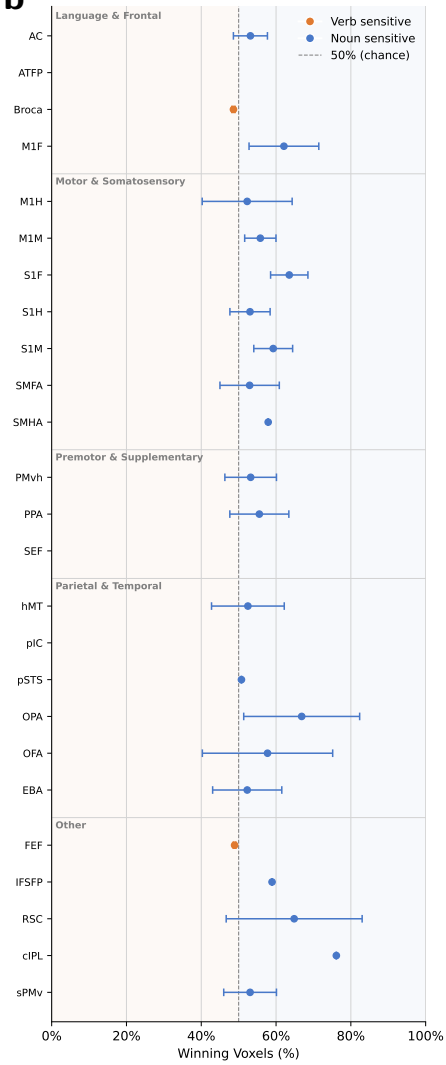


Figure 3. Test caption for Fig. 3, describing the individual subject results for the listening modality.

a ROI Dominant Category (mean $\pm$ SD) — listening (n=8)



b ROI Dominant Category (mean $\pm$ SD) — reading (n=8)



4 Discussion

		AC	EVC	VTC	LTC	LPC	MPC	IPC	SPC	BA
Nouns	Fronto-Temporal	–	–	↑	↑	–	–	↓	–	↓
	Object-Action (Semantic)	–	↑	↑	↑	–	–	–	–	–
	Combinatorial (Morphological)	–	–	–	–	–	–	–	–	–
	Lexicalist (Lemma-based)	–	–	–	↑	–	–	–	–	–
	Distributed Account	↑	↑	↑	↑	↑	↑	↑	↑	↑
Verbs	Fronto-Temporal	–	–	↓	↓	–	–	↑	–	↑
	Object-Action (Semantic)	–	–	–	–	↑	↑	↑	–	↑
	Combinatorial (Morphological)	–	–	–	–	–	–	↑	↑	↑
	Lexicalist (Lemma-based)	–	–	–	↑	–	–	–	–	–
	Distributed Account	↑	↑	↑	↑	↑	↑	↑	↑	↑

Table 1: Predicted brain area response to noun and verb processing according to different psycholinguistic models and the observed results in this thesis. The arrows indicate whether the observed response was higher (↑) or lower (↓) compared to the baseline model, while dashes indicate no significant difference.

5 Literature

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